

Introduction to Data Analysis

Instructor:
Seba Mostafa

ABOUT ME



Seba Mostafa
Data Analyst

I am Sepa, I have a BSc in Science and have been working as a senior data analyst and data analytics trainer for the past 5 years.

I have been working with both global and local companies (Udacity, Orange, Afri Labs, Carerha, ...).

- Published a paper on machine learning in international journal.
- Trained over 1000 Students.
- Students who graduated got job offers.
- Udacity Top Rated Instructor.
- Over 1000 hours training hours done.



Name:
College:
Field Of Work / Specialization:
Your Goal:
Your Expectation:



Agenda

- Data Analysis VS Data Science
- Fields Data Analysis
- Skills
 - Technical Skills
 - Soft Skills & Mentality
- Types Of Data Analytics
- Data Analysis as a Role
- Steps When Approaching New Project
- Tools Needed for Data Analysts
- Course Content

What is Data Analysis?



Introduction:

Data analytics, also known as data analysis, is a crucial component of modern business operations.

It involves examining datasets to uncover useful information that can be used to make informed decisions. This process is used across industries to optimize performance, improve decision-making, and gain a competitive edge.

Data Definition:

Data is a collection of facts, such as numbers, words, measurements, observations, or descriptions of things.

Types of Data:

Qualitative Data (Categorical):

Nominal: Egypt , USA , Germany

Ordinal: Good, Very Good, Excellent

Quantitative Data (Numerical):

**Discrete: Ex: No of Students
(23,24)**

**Continuous: Ex Sales, bank
balance (10.05)**

Applications of Data Analysis

1. Faster and better business decision
2. Predict future sales and purchases
3. Analyze the effectiveness of marketing campaigns



In short ...

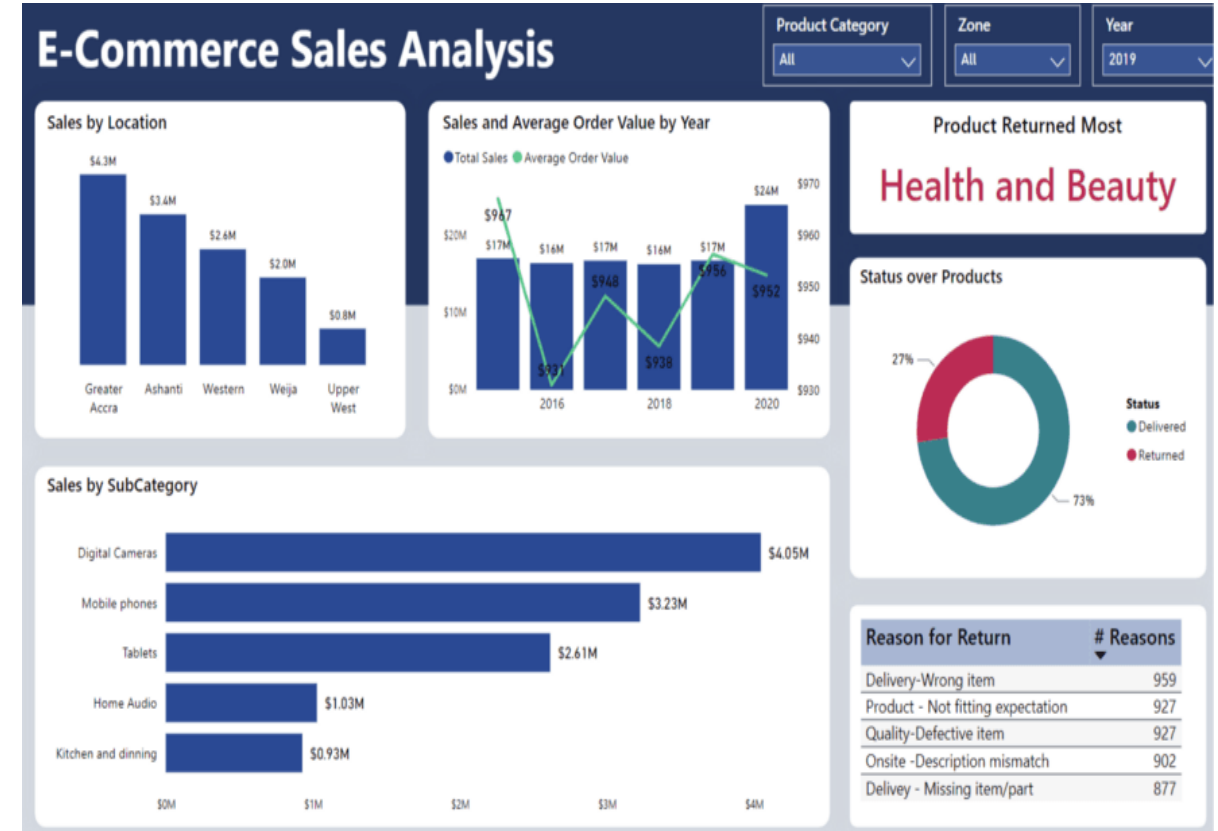
Data analyst should
be the decision
maker

Based on some historical
data, the analyst knew very
important information about
a specific product so that
could increase sales

Companies collect vast amounts of data, but in its raw form.
When data is in this raw form, it doesn't have any meaning or uses

Gourmet Hot Dog Shoppe - Microsoft Excel

2010										
2004										
Qtr1	Qtr2	Qtr3	Qtr4	Total	2005					
Qtr1	Qtr2	Qtr3	Qtr4	Total	Qtr1	Qtr2	Qtr3	Qtr4	Total	
\$838	\$1,117	\$906	\$739	\$3,600	\$1,338	\$1,450	\$1,157	\$1,239	\$5,184	
\$541	\$820	\$609	\$442	\$2,412	\$1,041	\$1,153	\$860	\$942	\$3,996	
\$497	\$776	\$565	\$398	\$2,236	\$997	\$1,109	\$816	\$898	\$3,820	
\$508	\$787	\$576	\$409	\$2,280	\$1,008	\$1,120	\$827	\$909	\$3,864	
\$519	\$798	\$587	\$420	\$2,324	\$1,019	\$1,131	\$838	\$920	\$3,908	
\$530	\$809	\$598	\$431	\$2,368	\$1,030	\$1,142	\$849	\$931	\$3,952	
\$662	\$941	\$730	\$563	\$2,896	\$1,162	\$1,274	\$981	\$1,063	\$4,480	
\$673	\$952	\$741	\$574	\$2,940	\$1,173	\$1,285	\$992	\$1,074	\$4,524	
\$783	\$1,062	\$851	\$684	\$3,380	\$1,283	\$1,395	\$1,102	\$1,184	\$4,964	
\$816	\$1,095	\$884	\$717	\$3,512	\$1,316	\$1,428	\$1,135	\$1,217	\$5,096	
\$827	\$1,106	\$895	\$728	\$3,556	\$1,327	\$1,439	\$1,146	\$1,228	\$5,140	
\$442	\$721	\$510	\$343	\$2,016	\$942	\$1,054	\$761	\$843	\$3,600	
\$563	\$842	\$631	\$464	\$2,500	\$1,063	\$1,175	\$882	\$964	\$4,084	
\$574	\$853	\$642	\$475	\$2,544	\$1,074	\$1,186	\$893	\$975	\$4,128	
\$552	\$831	\$620	\$453	\$2,456	\$1,052	\$1,164	\$871	\$953	\$4,040	
\$354	\$633	\$422	\$255	\$1,664	\$854	\$966	\$673	\$755	\$3,248	
\$365	\$644	\$433	\$266	\$1,708	\$865	\$977	\$684	\$766	\$3,292	
\$464	\$743	\$532	\$365	\$2,104	\$964	\$1,076	\$783	\$865	\$3,688	
\$475	\$754	\$543	\$376	\$2,148	\$975	\$1,087	\$794	\$876	\$3,732	
\$486	\$765	\$554	\$387	\$2,192	\$986	\$1,098	\$805	\$887	\$3,776	
\$409	\$688	\$477	\$310	\$1,884	\$909	\$1,021	\$728	\$810	\$3,468	
\$420	\$699	\$488	\$321	\$1,928	\$920	\$1,032	\$739	\$821	\$3,512	
\$596	\$875	\$664	\$497	\$2,632	\$1,096	\$1,208	\$915	\$997	\$4,216	



This report can tell a story and it's easily understood.

Data Analysis VS Data Science

Data Analysis

VS

Data Science



Fields of Data Analysis

Data analysis is found in all fields

Sales

Marketing

Medical

Business Operations

Pharmaceutical

Real Estate



Skills

Technical Skills



Soft Skills & Mentality



Soft Skills & Mentality



Communication Skills



Presentation Skills



Decision Making



Teamwork



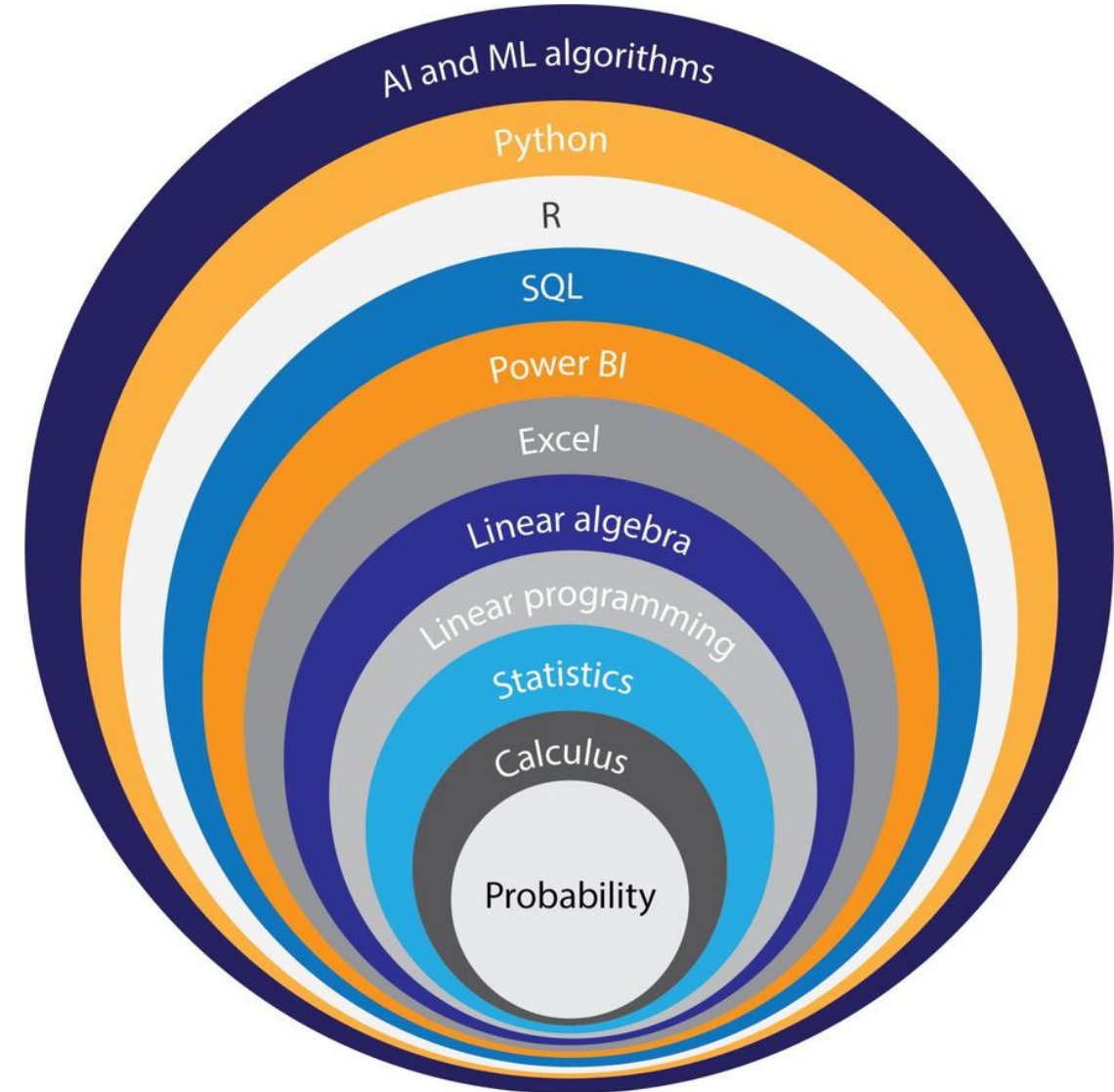
Problem Solving



Mentality and ways of thinking



Technical Skills



Types of Data Analytics

What do we need from the Data?

- 1 - What happened ?
- 2 - Why did it happen?
- 3 - What will happen ?
- 4 - What should we do ?

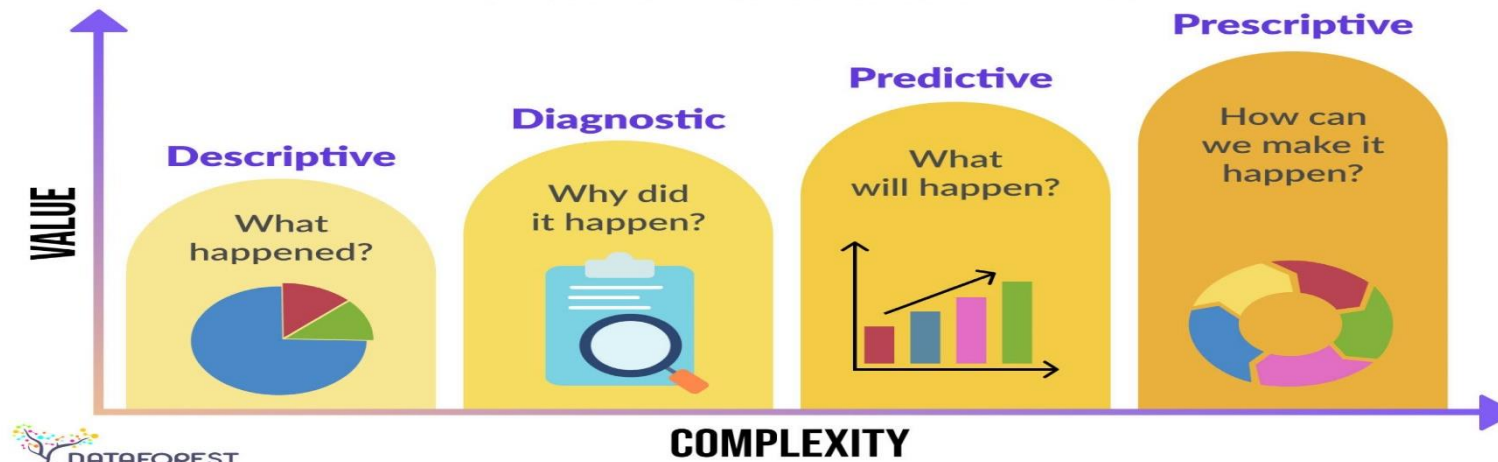
TYPES OF DATA ANALYTICS



In short ...

- Descriptive analysis (التحليل الوصفي) answers the question, “what happened?”
- Diagnostic analysis (التحليل التشخيصي) answers the question, “why did it happen?”
- Predictive analysis (التحليل التنبؤي) answers the question, “what might happen in the future?”
- Prescriptive analysis (التحليل الوصفي) answers the question, “what should we do about it?”

TYPES OF DATA ANALYTICS



Descriptive Data Analytics:

Descriptive analysis tells us what happened. It helps describe or summarize quantitative data by presenting statistics. For example, descriptive statistical analysis could show the distribution of sales across a group of employees and the average sales figure per employee.

Diagnostic Data Analytics:

If the descriptive analysis determines the “what,” diagnostic analysis determines the “why.” Let’s say a descriptive analysis shows an unusual influx of patients in a hospital, drilling into the data further might reveal that many of these patients shared symptoms of a particular virus. This diagnostic analysis can help you determine that an infectious agent—the “why”—led to the influx of patients.

Predictive Data Analytics:

So far, we've looked at types of analysis that examine and draw conclusions about the past. Predictive analytics uses data to form projections about the future. Using predictive analysis, you might notice that a given product has had its best sales during the months of September and October each year, leading you to predict a similar high point during the upcoming year.

Prescriptive Data Analytics:

Prescriptive analysis takes all the insights gathered from the first three types of analysis and uses them to form recommendations for how a company should act. Continuing with our previous example, this type of analysis might suggest a marketing plan to build on the success of the high sales months and harness new growth opportunities during the slower months.

Data Analysis as a Role



1. Domain Knowledge

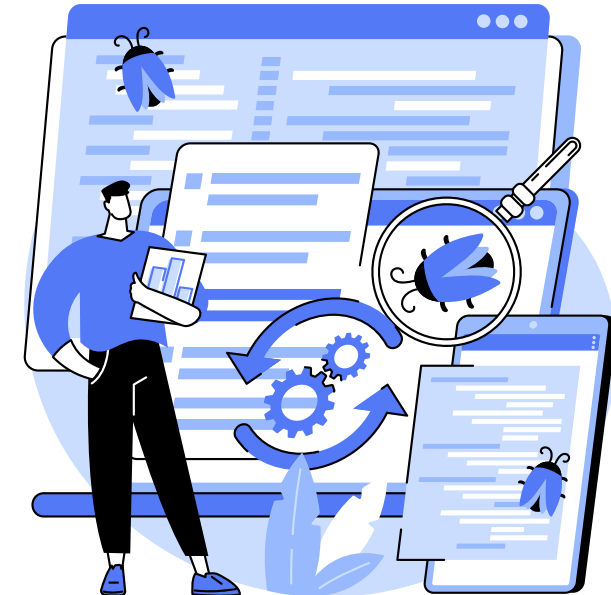
Domain knowledge is the most important

You can never solve a problem that you do not understand

So, you must understand the business that you are solving.

Ex: We have a problem with marketing:

We should understand the product need and the client to whom we are delivering the product so we can ask for data and produce a solution.



2. Gather Relevant Data

Getting the data needed

After understanding the business problem we're trying to solve, we then start getting the data that can support the analysis of

- ❖ How the problem happened ?
- ❖ How long it may last ?
- ❖ How to Solve the problem ?



3. Define **data quality** and implement **data preprocessing**

We cannot perform accurate analysis with poor-quality data

So, we must define data quality standards and clean the data accordingly

For example, we need to:

- ❖ Remove duplicate entries
- ❖ Handle missing values
- ❖ Standardize data formats

This helps in producing reliable and valid results from the analysis.



4. Report the results using data visualization tools

We cannot convey insights effectively without proper visualization

So, we must use data visualization tools to report the results

For example, we can use:

- ❖ Charts and graphs
- ❖ Dashboards
- ❖ Interactive visualizations

This helps stakeholders understand the analysis and make informed decisions



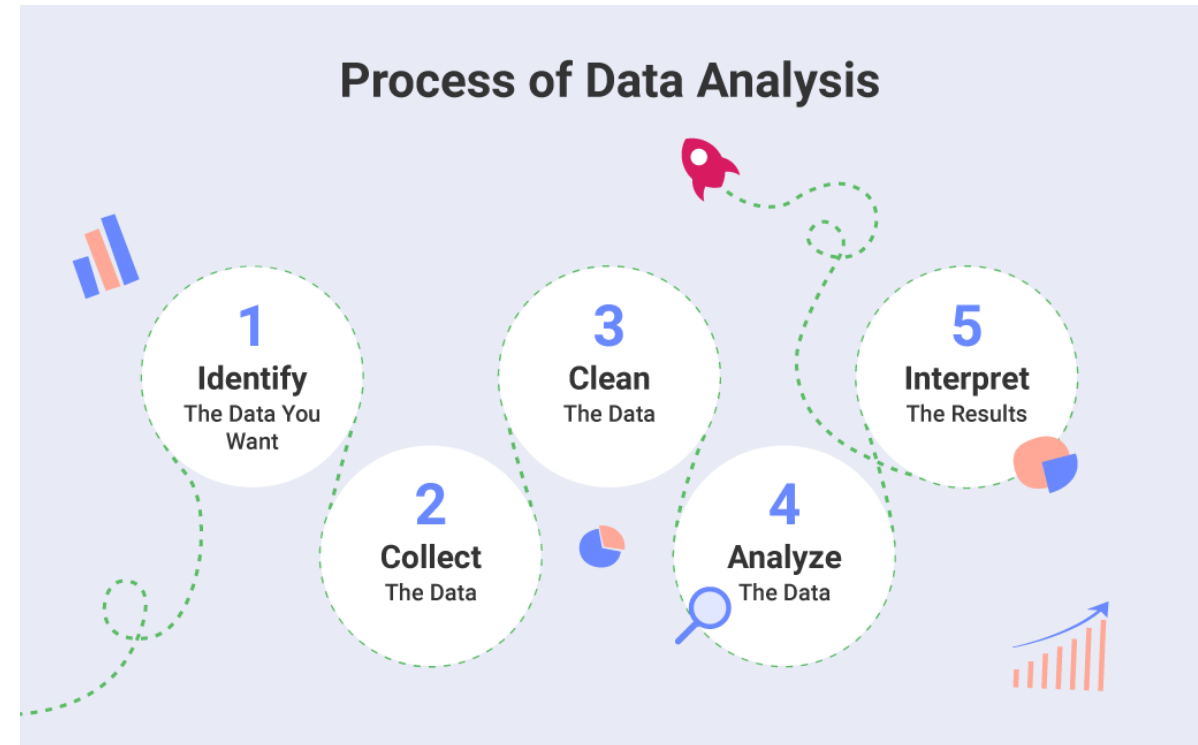
Now let's assume that a data analyst is approaching a new project..

What are the steps needed?

In short ...

Process of Data Analytics contains 4 steps:

1. Data Collection
2. Data Cleaning
3. Data Analysis And Data Interpretation
4. Data Visualization



1. Define the questions you need to answer



2. Collect the data



Where do we get the data?

The data can be collected through multiple resources

1.Databases

2.Websites

3.Or even manually

Whatever the way, there must be some data that is related to my problem.

3. Clean the data



What to clean?

The data might have some problems
like

1. Anomalies (Outliers)

2. Duplicates

3. Missing data

etc.

4. Analyze the data



Reporting

This step is about putting your final results in a nice looking and easy to understand dashboard or dynamic report.

Here you give some answers to the questions

Tools Needed for Data Analysts

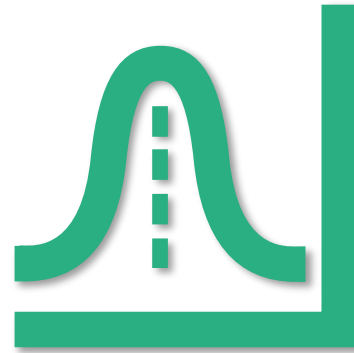
1. Programming Skills



Python programming language

Python is mostly used for data preparation and quick analysis

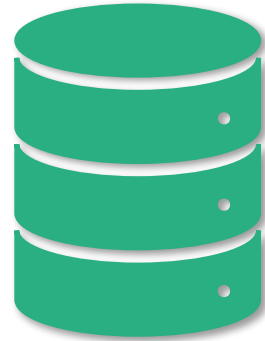
2. Math and Statistics



3. Domain Knowledge



4. Database and SQL

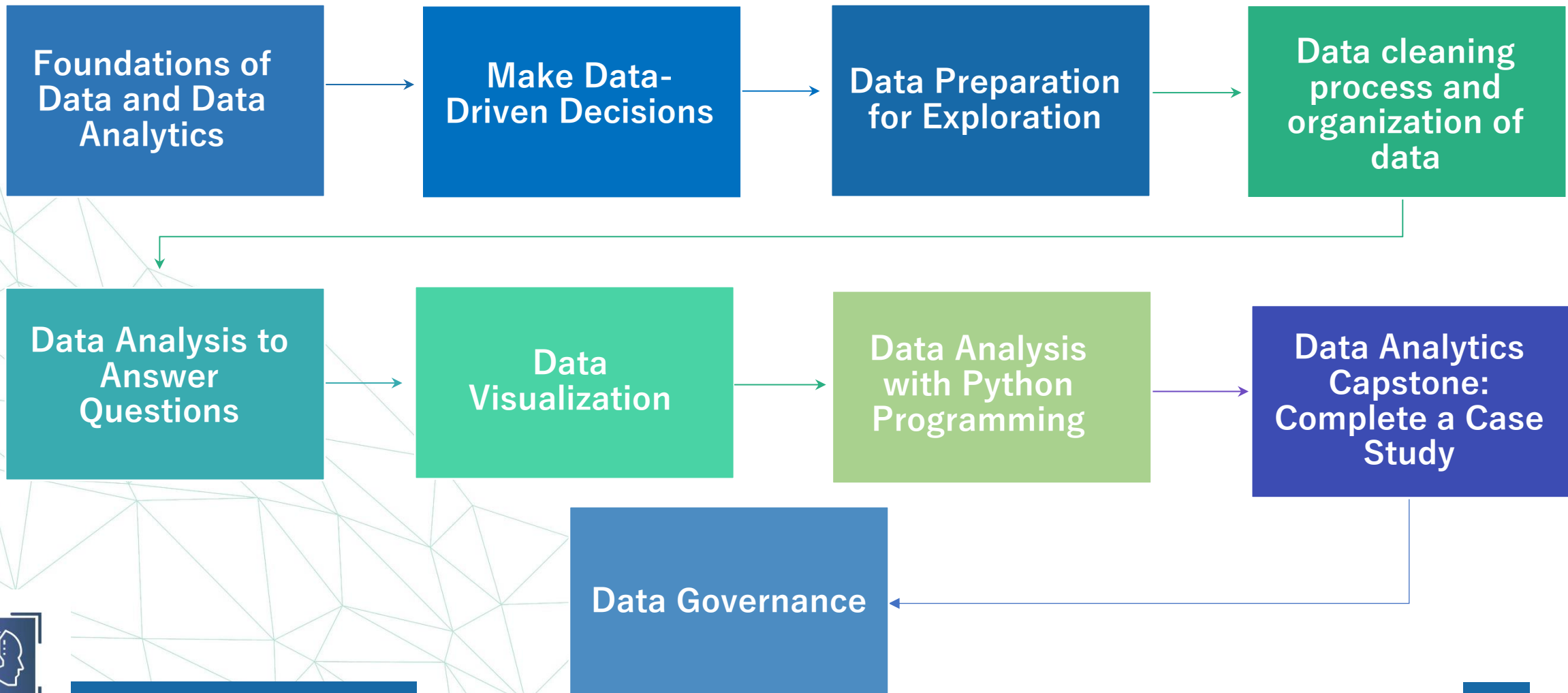


5. Visualization and Reporting Tools



Course Content

Course Structure





Any Questions ?



**WE WANT YOUR
FEEDBACK**

Thank you!

